



1 RAPPORTAGE: NULINSPECTIE & EMC GAP-ANALYSE

Project: Simonshaven 380kV Uitbreiding (DP2-4)

Client: SC&M II V.O.F.

DNV Reference: OPP-00436810-PSP 25-1757

Date: xx-12-2025

Status: DRAFT (Pending Site Visit & DP2-4 DO Documentation)

2 MANAGEMENT SUMMARY

This report documents the findings of the EMC Baseline Survey (*Nulinspectie*) performed by DNV at the Simonshaven 380kV substation.

Objective: To visually verify the "As-Built" EMC status of the existing Phase 1 (DP1) installation and identify specific integration risks for the new Phase 2-4 (DP2-4) expansion.

Key Conclusion (Preliminary):

Based on the review of the DP1 As-Built documentation [2], it is a known fact that the existing buildings (CDG and Takhuisjes) **do not** fully comply with the TenneT *Standard 380kV Zone Concept* [1] regarding building shielding (LPZ 2). This inspection serves to confirm the extent of these deviations. **Critical Note:** If the existing buildings are confirmed to be unshielded (LPZ 1), the design for DP2-4 might be subject to higher immunity specifications or cabinet-level shielding for all new secondary equipment installed in these rooms. This will be evaluated during the EMC management design.

3 INTRODUCTION

3.1 Project Context

SC&M is executing the expansion of the Simonshaven substation (DP2-4). This involves integrating new high-voltage assets (Transformers TR412-414, Compensation Coils, Bays 4-5) into the existing operational environment (DP1).

3.2 Scope of Inspection

The scope of this *Nulinspectie* is limited to:

1. **Visual Inspection:** Verification of visible earthing, bonding, shielding, and cable routing in the existing CDG, Takhuisjes, and accessible cable trenches.
2. **Gap Analysis:** Comparing the physical installation against TenneT Standard E04 [1] and the DP1 EMC Plan [2].
3. **Interface Check:** Identifying physical connection points for the new DP2-4 earthing and cabling.

Exclusion: This scope does not include electromagnetic field measurements, soil resistivity measurements, or inspection of covered/underground components unless opened by the client.

3.3 Dependencies & Limitations

- **Missing Design Data:** At the time of this inspection, the **Definitief Ontwerp (DO)** for DP2-4 has not been provided. DNV cannot assess specific cable routing conflicts for the new work until these documents are received.
- **Hidden Defects:** DNV assumes that underground earthing connections (not visible) were installed according to the as-built drawings, unless visual evidence suggests otherwise.

4 REFERENCE FRAMEWORK

The inspection evaluates the site against the following hierarchy of documents:

Ref	Document Number	Title	Status	Relevance
[1]	1287698 (E04)	<i>Uitwerking EMC eisen-zone concept standaard 380kV station</i>	The Standard	TenneT's mandatory rules for zoning (LPZ) and installation details.
[2]	2407-5582	<i>EMC beheersplan SMH380 (Phase 1)</i>	As-Built	Describes the executed state of DP1, including known deviations.
[3]	01350-01-00075	<i>Beoordeling EMC studie SMH380 Fase 1</i>	History	Previous gap analysis highlighting missing shielding mesh.

4.1 Risk Identification (Pre-Inspection Analysis)

Based on the review of Document [2] (*EMC beheersplan SMH380*), the following specific risks must be verified on-site:

1. **Legacy Shielding Defect:** Document [2] states that the EMC mesh (30x30mm) in the CDG and Takhuisjes was "not correctly installed."
 - *Risk:* The rooms may effectively be **LPZ 1** (Unshielded) instead of **LPZ 2**.
 - *Inspection Focus:* Verify if any wall/floor bonding points for the mesh are visible or connected.

2. **Cable Entry Integrity:** There is ambiguity regarding the bonding of Roxtec frames and the termination of cable shields at the building entry.
 - *Risk:* High-frequency interference entering the building via cable shields.
 - *Inspection Focus:* Check physical bonding of frames to the earth bar (Detail 27 of [1]).
3. **PEC Continuity:** It is unconfirmed if Parallel Earthing Conductors (PECs) were installed in all concrete trenches.
 - *Risk:* Lack of shielding for field cabling.
 - *Inspection Focus:* Spot-check open trenches for 50mm² Cu PECs.

5 INSPECTION FINDINGS

This section is to be populated by the DNV Engineer following the site visit.

This section details the observations made during the visual inspection of the Simonshaven 380kV substation (DP1 area). The inspection focused on verifying the "As-Built" status against the TenneT *EMC eisen-zone concept* (Document E04) and identifying deviations reported in the Phase 1 documentation.

5.1 Central Service Building (CDG)

5.1.1 Cable Entry Systems (Kabelinvoer)

- **Inspection Scope:** Visual check of the Roxtec cable transit frames in the AC/DC and Telecom rooms. Verification of frame bonding to the main earth bar and correct termination of cable shields (360° bonding) within the modules, in accordance with *TenneT Detail 27*.
- **Observations:**
 - The physical condition of the Roxtec frames was found to be **[Good / Fair / Poor]**.
 - The bonding connection between the transit frame and the room's ring earth (aardrail) was **[Visible and connected / Disconnected / Corroded / Missing]**.
 - Regarding cable shields: **[Cable shields appear to be stripped and clamped inside the modules / Cable shields are carried through without clamping / It was not possible to verify shield termination visually]**.
- **Photo Reference:** [Insert Photo ID, e.g., IMG_001, IMG_002]

5.1.2 Room Shielding & Equipotential Bonding

- **Inspection Scope:** Verification of the Faraday cage concept (LPZ 2). Inspection of bonding points between the wall reinforcement (wapening/EMC-net) and the earthing system, and bonding of metal door frames.
- **Observations:**
 - Visible connections to the building reinforcement or EMC mesh were **[Observed at regular intervals / Not observed / Only observed at corners]**.
 - Metal door frames **[Are / Are not]** bonded to the structure earth with short flexible straps (litze).
 - **[Optional: Note any penetrations through the walls that are not treated, e.g., drilled holes for new cables without glands.]**
- **Photo Reference:** [Insert Photo ID]

5.1.3 Cabinet Installation (Secundaire Kasten)

- **Inspection Scope:** Check of cabinet earthing and cable gland plates (wartelplaten) in the CDG server/telecom room.
- **Observations:**
 - Cabinets are earthed via **[Short flat copper litz wire / Long round yellow-green wire (Pigtail)]**.
 - Cable entry into cabinets is via **[Conductive gland plates with EMC glands / Open bottom / Plastic glands]**.
- **Photo Reference:** [Insert Photo ID]

5.2 Bay Houses (Takhuisjes)

5.2.1 Building Shielding Integrity

- **Inspection Scope:** Verification of the "Missing Mesh" deviation noted in previous reports. Check for visible earth bars connected to the wall structure.
- **Observations:**
 - The inspection [**Confirms / Could not confirm**] the presence of the 30x30mm EMC mesh described in the design.
 - Earth bars (aardrails) on the walls [**Are / Are not**] connected to the building reinforcement at the required intervals (max 5m).
- **Photo Reference:** [Insert Photo ID]

5.2.2 Secondary Equipment Earthing

- **Inspection Scope:** Check of protection relay cabinets and auxiliary power distribution boards.
- **Observations:**
 - Earthing of the cabinets to the room ring earth is [**Consistent / Inconsistent**].
 - [**Note any specific cabinets where the earthing wire is excessively long or coiled, creating high impedance.**]
- **Photo Reference:** [Insert Photo ID]

5.3 Field Infrastructure (Buitenterrein)

5.3.1 Concrete Cable Trenches (Kabelgoten)

- **Inspection Scope:** Covers were lifted at [**Insert Number**] locations to verify the presence of a Parallel Earthing Conductor (PEC) and cable segregation.
- **Observations:**
 - **Location 1 (Near CDG):** A copper PEC ($\geq 50\text{mm}^2$) [**Was / Was not**] visible running parallel to the cables.
 - **Location 2 (Near Tak 1):** A copper PEC [**Was / Was not**] visible.
 - **Cable Segregation:** High voltage (if any), Low voltage, and Signal cables [**Are separated by dividers / Are mixed in the trench**].
- **Photo Reference:** [Insert Photo ID]

5.3.2 Primary Equipment Bonding

- **Inspection Scope:** Visual check of earthing connections to steel portals and support structures (TenneT requirement: 2x connections for main structures).
- **Observations:**
 - Steel portals show [**Double / Single**] earthing connections to the main grid.
 - The condition of the connections is [**Good / Corroded / Loose**].
- **Photo Reference:** [Insert Photo ID]

5.3.3 Fence Earthing (Hekwerk)

- **Inspection Scope:** Verification of earthing straps between fence panels and connection to the station earth grid to prevent touch voltage hazards.
- **Observations:**

- Flexible earthing straps between fence panels are **[Present and intact / Missing in some sections / Broken]**.
- Connection to the main earth grid was **[Verified at intervals / Not visible]**.
- **Photo Reference:** [Insert Photo ID]

5.4 General Observations

- **Housekeeping:** The general level of EMC housekeeping (e.g., loose wire clippings, metal debris in cabinets) is **[Good / Needs Improvement]**.
- **Modifications:** There is evidence of recent cabling modifications where EMC glands were **[Used correctly / Replaced with plastic / Left open]**.

Consider integrating the below tables in the headings above.

5.5 Central Service Building (CDG) – 9:15 – 9:45

Item	Check	Finding / Observation	Photo Ref	Status
5.1.1	Cable Entry (AC/DC Room) Ref: Detail 27	[PLACEHOLDER: Describe state of Roxtec frames. Are they earthed to the ring earth?]		
5.1.2	Cable Entry (Telecom) Ref: Detail 27	[PLACEHOLDER: Are cable shields stripped and clamped <i>inside</i> the module?]		
5.1.3	Room Shielding Ref: Bijlage 2.5	[PLACEHOLDER: Are door frames bonded to the wall structure? Any visible mesh connections?]		

5.6 Bay Houses (Takhuisjes) – 9:50 – 10:10

Item	Check	Finding / Observation	Photo Ref	Status
5.2.1	Building Shielding Ref: Bijlage 2.1	[PLACEHOLDER: Confirm if the "missing mesh" deviation is visible. Check earth bars on walls.]		
5.2.2	Cabinet Earthing Ref: Bijlage 1	[PLACEHOLDER: Check gland plates. Are EMC glands used? Are pigtails used (NOK)?]		

5.7 Field Infrastructure – 10:15 – 11:00

Item	Check	Finding / Observation	Photo Ref	Status
5.3.1	Concrete Trenches Ref: Bijlage 4	[PLACEHOLDER: Lift cover. Is a copper PEC visible running parallel to signal cables?]		

5.3.2	Fence Earthing Ref: AM-Req-1256	[PLACEHOLDER: Check earthing straps between fence panels at the interface point for DP2-4.]		
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6 GAP ANALYSIS & RECOMMENDATIONS FOR DP2-4

To be finalized after inspection.

6.1 Confirmed Deviations

Based on the inspection, the following deviations from TenneT Standard [1] are confirmed in the existing DP1 installation:

- **Deviation 1:** [PLACEHOLDER e.g., CDG Building Shielding is ineffective.]
- **Deviation 2:** [PLACEHOLDER e.g., PEC missing in Trench Section X.]

6.2 Implications for DP2-4 Design

To ensure EMC compliance for the new expansion despite these legacy issues, DNV recommends the following strategy for the DP2-4 design:

1. **Zoning Strategy:** The EMC Management Plan for DP2-4 will classify the existing buildings as **LPZ 1**.
2. **Mitigation:** New secondary equipment for DP2-4 installed in these buildings must rely on **Cabinet-Level Shielding** (Metal housings, EMC glands) rather than Room-Level shielding.
3. **Retrofit (If required):** [PLACEHOLDER e.g., "The earth bar in Takhuisje 3 must be cleaned/tightened before connecting new DP2-4 cables."]

7 APPENDIX A: SITE INSPECTION CHECKLIST

Field use only.

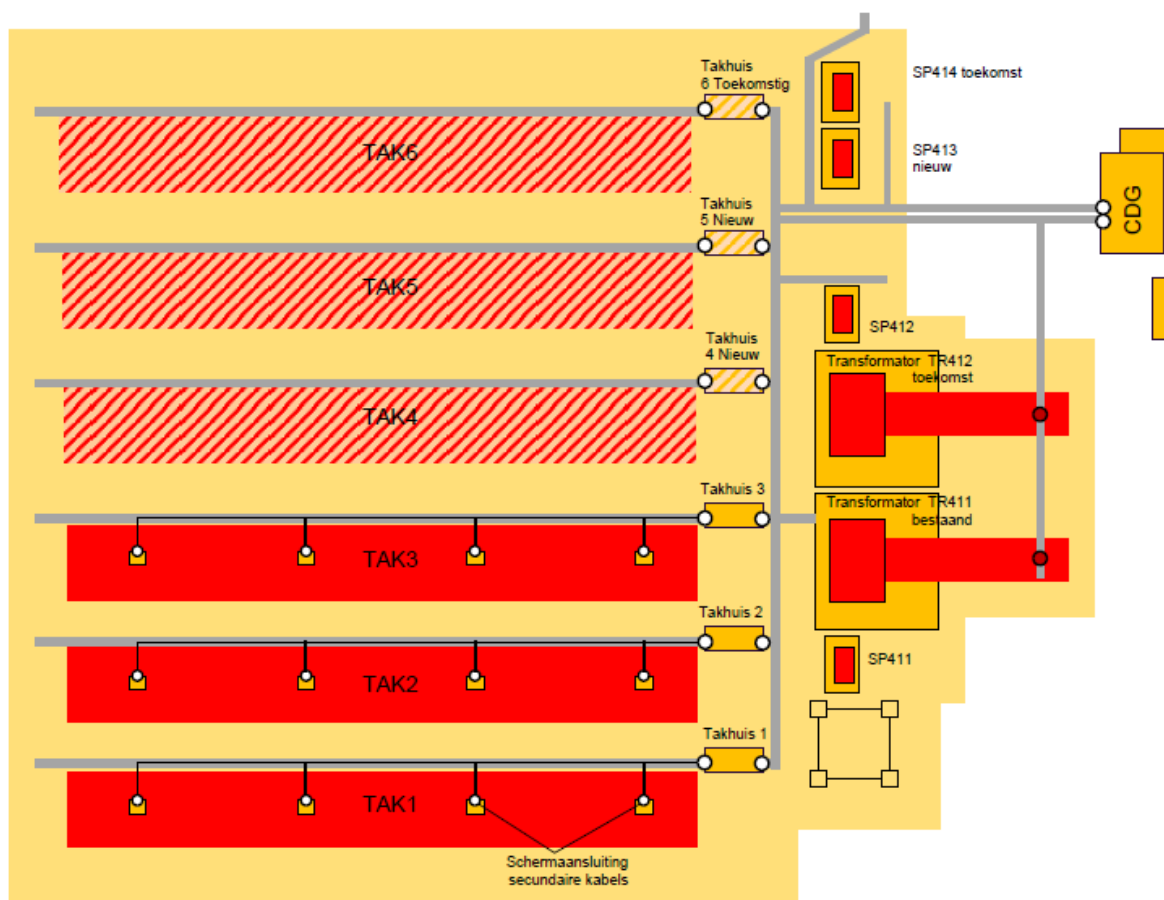
ID	Asset	TenneT Ref (E04)	Criteria	OK / NOK	Notes
A1	CDG Entry	Detail 27	Frame bonded to Earth Bar?		
A2	CDG Entry	Detail 27	Shields clamped in module?		
A3	Takhuisje	Bijlage 2.1	Wall reinforcement bonded?		
A4	Cabinets	Bijlage 1	EMC Gland Plate present?		
A5	Cabinets	Foto 9963	No long "Pigtails" on screens?		
A6	Trenches	Bijlage 4	PEC (Cu) visible in trench?		
A7	Fence	AM- Req- 1256	Earthing straps present?		



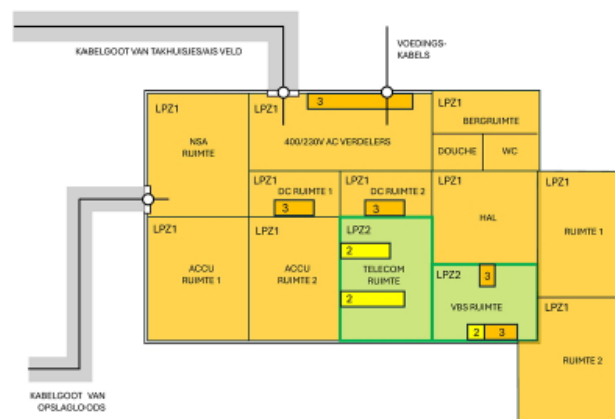
Bijlage A EM zone indeling SMH380

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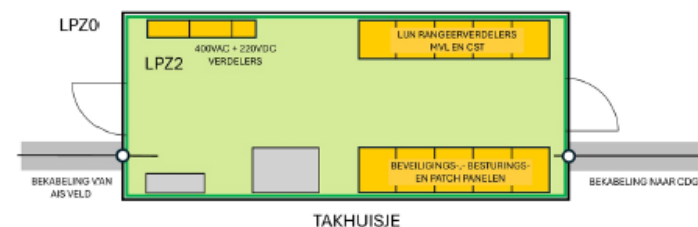
- 2** = Apparatuur in takhuisjes en het CDG die beschermd wordt door extra afscherming, apparatuur in telecomruimte die geen industriële immuïteitsspecificaties heeft.
- 3** = industrieel, secundaire installatie in takhuisjes en CDG en langs hekwerken rond de AIS schakeltuin.
- 4** = locaties binnen de primaire schakeltuin, ter plaatse van railsystemen, schakelaars, scheiders, transformatoren, spoelen en kabeleindsluitingen.
- = EM/LPZ zone overgang d.m.v. kabelscherm aansluiting secundaire kabels bij invoer veldkasten, takhuisjes en CDG
- = Eindsluitingen HS kabels



Figuur A.1 EM zone indeling SMH380



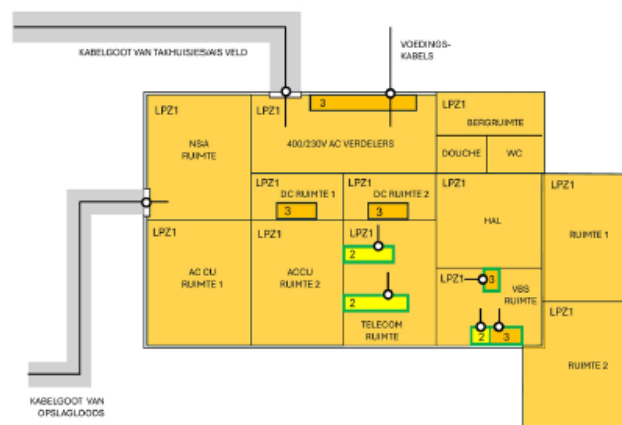
EM zonering CDG conform TenneT eisen



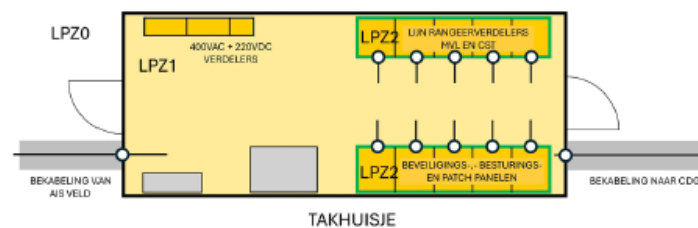
EM zonering takhuisje conform TenneT eisen

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-  = LPZ1, afscherming door middel van betonwapening overeenkomend met EM3
-  = LPZ2, afscherming door middel van EMC-net 30x30x3mm
-  = EM2 apparatuur
-  = EMC kabeldoorvoer met schermaansluiting



EM zonering CDG conform zoals uitgevoerd



EM zonering takhuisje conform zoals uitgevoerd

Figuur A.2 EM en LPZ zone indeling CDG en takhuisjes